

Meeting Status

Automatic presence sign on e-paper via Bluetooth
User Manual



The tag shows your status in real time – here "IN MEETING".

Software developed by SA7BNB, Anders Isaksson

This program automatically detects when you are in a meeting or call on your computer and shows your status – **AVAILABLE** or **IN MEETING** – both as a colored dot in the Windows system tray and on a small Bluetooth-connected e-paper sign that you can place outside your office or on your monitor.

1. Overview

Meeting Status is a small background program for Windows. It monitors whether any app is using your microphone or camera – which happens as soon as you join a meeting in Teams, Zoom, Meet, Webex, Slack, Discord or similar – and updates your status accordingly.

The status is shown in two places:

- **In the system tray:** a green dot when you are available, red when you are busy.
- **On the e-paper tag:** your name, your status, an optional icon and a colored frame around the whole sign.

Two status modes



AVAILABLE – black text and black frame



IN MEETING – red text and red frame

Why e-paper? The screen keeps its image with no power at all and only draws battery when it is updated. That is why the tag is **only written when your status actually changes** – not continuously. One update takes 15–20 seconds, which is normal for e-paper.

2. Installation

What you need

- A Windows PC with Bluetooth (most laptops have it built in).
- A Gicisky / PICKSMART 2.9" e-paper tag (black/white/red), which advertises with the name NEMRxxxxxxx.
- The program itself (the .exe or the Python script).

If you have the .exe

Just double-click **MeetingStatus.exe**. Nothing else is needed – the settings file, the log and the help file are created automatically next to the program the first time it runs.

If you run the Python script

Install the dependencies and start it:

```
pip install pystray Pillow bleak
python MeetingStatus.py
```

Build the .exe yourself

To turn the script into a standalone executable:

```
pip install pystray Pillow bleak pyinstaller
python -m PyInstaller --onefile --noconsole --name MeetingStatus ^
    --hidden-import=PIL._tkinter_finder ^
    --collect-all pystray --collect-all PIL --collect-all bleak ^
    MeetingStatus.py
```

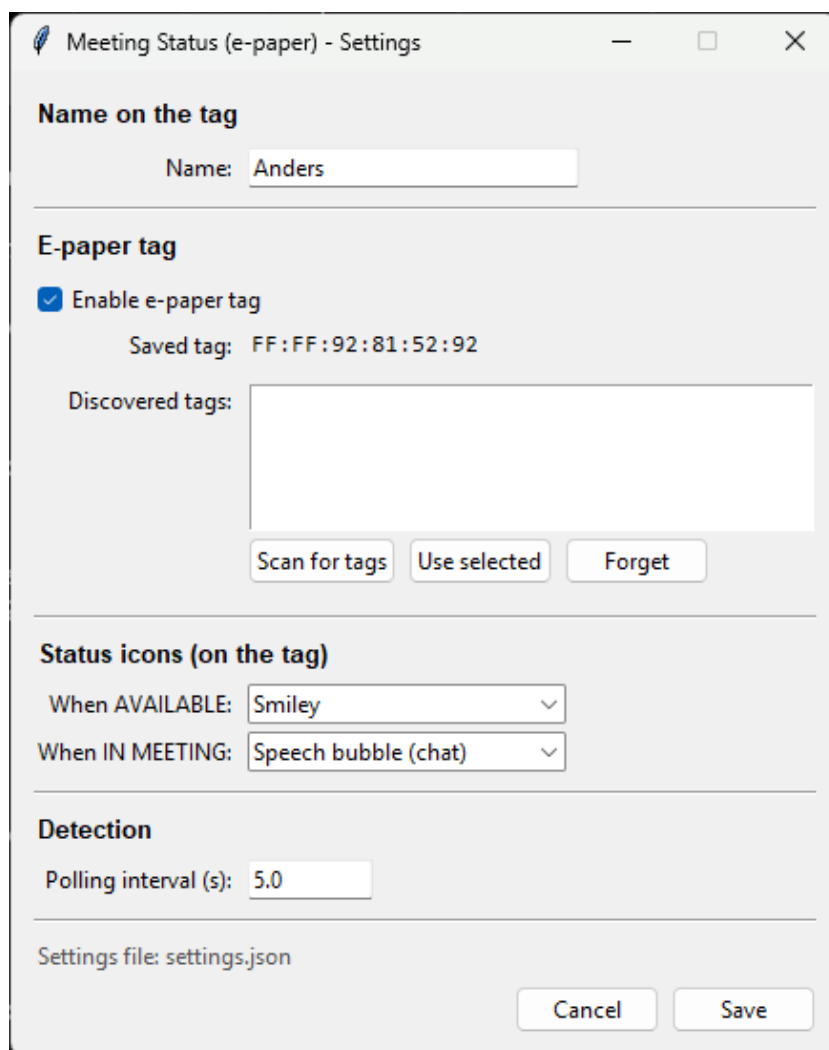
The result is **dist\MeetingStatus.exe**.

Start automatically at login

Press **Win+R**, type **shell:startup** and press Enter. Place a shortcut to the program in the folder that opens, and it will start every time you log in.

3. First run and settings

The first time you start the program, the settings window opens automatically. You can reach it at any time by right-clicking the icon in the system tray and choosing **Settings...**



The settings window.

Steps to pair the tag

1. Wake the tag by removing and reinserting a battery (it only advertises for a short while after waking).
2. Check **Enable e-paper tag**.
3. Click **Scan for tags** and wait a few seconds.
4. Select your tag in the list (e.g. NEMR92815292) and click **Use selected**.
5. Type your name in the **Name** field – it appears at the top of the tag.
6. Choose optional icons for AVAILABLE and IN MEETING.
7. Click **Save**.

The program remembers the tag's address and reconnects automatically next time. If the scan finds no tag, wake it with a battery swap and scan immediately.

4. Settings in detail

Setting	Description
Name	The name shown at the top of the tag.
Enable e-paper tag	Turns writing to the tag on or off. The tray dot works regardless.
Saved tag	The saved Bluetooth address (e.g. FF:FF:92:81:52:92).
Scan / Use selected / Forget	Scan for tags, select one from the list, or forget the saved one.
When AVAILABLE	Icon shown next to "AVAILABLE". Can be (none).
When IN MEETING	Icon shown next to "IN MEETING". Can be (none).
Polling interval (s)	How often the program checks the mic/camera (default 5 seconds).

Icons

The icons are drawn as clean single-color symbols that suit e-paper. You freely choose which one is shown for each status – or none at all.

Icons for **AVAILABLE**:



Check



Thumb



Coffee



Dot



Smiley

Icons for **IN MEETING**:



Speech



Camera



Headset



Phone



Video

5. The system tray menu

Right-click the program icon in the Windows system tray to reach the menu.

Status	Current mode: AVAILABLE, IN MEETING or OFF.
Source	Which app triggered "busy" (e.g. Teams.exe).
Tag	The tag's state: ready, writing... or off.
Settings...	Opens the settings window.
Update tag now	Forces an immediate redraw of the tag.
Setup help	Opens the built-in help text.
Quit	Exits the program.

About mute

When you click "mute" in Teams or similar, the microphone is still kept open at the system level – the app just stops sending audio. The program therefore shows you as **IN MEETING** even while muted. The only mode that is not detected is when you are muted, the camera is off, and you are only listening.

6. Troubleshooting

The tag does not update

- Check that the tag is awake and within range (~10 m).
- Make sure the correct address is saved in the settings.
- Look in the log file `meeting_status.log` next to the program.

The scan finds no tags

- Tags only advertise for a short while after waking – remove and reinsert a battery, then scan immediately.
- Check that Bluetooth is turned on in Windows.
- Some desktop PCs have no Bluetooth – use a USB dongle or a laptop.

The text on the tag looks wrong (rotated, mirrored or wrong color)

- Adjust the PANEL constants at the top of the script: `PANEL_ROTATION`, `PANEL_INVERT_BW` and `PANEL_INVERT_RED`.

Windows Defender or antivirus flags the .exe

- This is a common false positive for PyInstaller-built programs. Approve the file manually if you trust it.

Tag model: Gicisky / PICKSMART 2.9" e-paper, 296×128 pixels, black/white/red, Bluetooth Low Energy.

Purchased from: AliExpress – "2.9" E Ink Screen Price Tag".

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Windows only. Use at your own risk.